

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 8 - Unit Test 1

Unit Test 1 - Answer Key

Class: Class 8

Assessment Type: Unit Test 1

Total Marks: 10

Duration: 30 minutes

Topics Covered: Functions

ICT Teaching Resources Portal

Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Section I: Multiple Choice Questions (5 x 1 = 5 marks)

Q.No	Answer	Explanation
1	b	def keyword is used to define functions in Python
2	c	TypeError because add requires 2 arguments but only 1 was provided
3	a	lambda x: x * 2 creates a lambda function
4	b	Default parameter "World" is used when no argument is passed
5	c	return keyword sends a value back from the function

Section II: Short Answer Questions (5 x 1 = 5 marks)

1. Write a Python function `power(base, exp)`.

• **Answer:**

```
def power(base, exp):  
    result = 1  
    for i in range(exp):  
        result *= base  
    return result  
  
print(power(2, 5)) # Output: 32
```

• **Marking:** 1 mark for correct function definition, 1 mark for correct output.

2. What is the output of the global variable modification?

• **Answer:** Output: 15 and 15. The `global` keyword allows the function to modify the global variable `x`. Inside the function, `x` becomes 15, and this change persists outside.

• **Marking:** 1 mark for correct output, 1 mark for explanation.

3. Differentiate between recursion and iteration.

• **Answer:** Recursion is when a function calls itself (e.g., factorial function). Iteration uses loops like `for` or `while` to repeat code (e.g., counting from 1 to 10). Recursion uses stack memory; iteration uses constant memory.

• **Marking:** 1 mark for explaining recursion with example, 1 mark for explaining iteration with example.

4. Write the output of the mystery function.

• **Answer:** "OLLEH" and "NOHTYP". The function reverses the string using slice notation `[::-1]`.

• **Marking:** 1 mark for each correct output.

5. Explain what happens when a function has no return statement.

• **Answer:** When a function has no return statement, it returns `None` by default. Example: `def greet(): print("Hello")` returns `None` when called. To get a value back, you must use `return`.

• **Marking:** 1 mark for explaining default return value, 1 mark for example.